Introduction to Cryptography

Lecture 43: Firewalls, Cybercrime, Intellectual Property, Privacy and Ethics

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Source coverage note

The provided extract does not cover firewalls, cybercrime, intellectual property, privacy, or ethics. This lecture is supplementary material aligned to the requested syllabus.

Learning objectives

- Explain firewall functions and architectures
- Classify common cybercrime scenarios
- Differentiate major forms of intellectual property
- Discuss privacy and professional ethics

Firewall concept

- Enforces a security policy on network traffic
- Inspects connections crossing a trust boundary
- One layer of defense, not a complete security solution

Firewall policy

- Default-deny versus default-allow
- Allow only required services
- Document and periodically review rules

Packet-filtering firewall

- Evaluates headers such as IP addresses, ports and protocols
- Fast and straightforward
- Limited awareness of application content

Stateful inspection

- Tracks established connection state
- Uses context beyond an isolated packet
- Can reject unexpected traffic

Application proxy firewall

- Intermediates selected application protocols
- May inspect application requests
- Adds deployment and performance considerations

Next-generation firewall

- May combine stateful filtering and application awareness
- Can integrate intrusion-prevention features
- Capabilities depend on product and configuration

Firewall placement

- Perimeter boundary
- Internal segmentation
- Host-based controls
- Cloud security groups and virtual network policies

DMZ concept

- Separate public-facing services from internal systems
- Apply distinct access rules for each zone
- Reduce direct exposure of sensitive systems

Firewall limitations

- Cannot correct vulnerable application code
- Cannot stop every insider misuse
- Encrypted traffic may limit inspection
- Poor rules can defeat intended protections

Firewall design exercise

- Web server requires HTTPS from the internet
- Database should accept connections only from web tier
- Draft a least-privilege rule set

Cybercrime overview

- Unlawful acts involving computers, networks or data
- Computer may be target, tool or evidence
- Legal definitions vary by jurisdiction

Cybercrime examples

- Unauthorized access
- Online fraud and impersonation
- Data theft and extortion
- Service disruption

Social engineering

- Manipulation of human trust
- Phishing and impersonation
- Verify requests using trusted channels

Digital evidence

- Preserve logs and relevant artifacts
- Maintain integrity and chain of custody
- Follow applicable legal and institutional procedures

Intellectual property (IP)

- Legal rights associated with creative and technical work
- Different protections cover different subject matter
- Rights and exceptions vary by jurisdiction

Copyright

- Protects original expression such as software code and written works
- Does not generally protect an abstract idea itself
- Licensing determines permitted reuse

Patents

- May protect qualifying inventions
- Typically require novelty and other legal criteria
- Disclosure and filing rules depend on jurisdiction

Trademarks and trade secrets

- Trademark: identifies source of goods or services
- Trade secret: commercially valuable confidential information
- Protection depends on distinct legal requirements

Software and academic integrity

- Respect open-source license conditions
- Attribute borrowed material appropriately
- Avoid plagiarism and unauthorized distribution

Privacy fundamentals

- Collect only data needed for a stated purpose
- Provide transparency and appropriate access controls
- Retain data only as justified

Privacy versus security

- Security protects confidentiality, integrity and availability
- Privacy concerns appropriate processing of personal information
- A system can be secure yet still invade privacy

Data protection in practice

- Minimize collection
- Limit access
- Use encryption where appropriate
- Set retention and deletion procedures

Ethical responsibilities

- Obtain authorization before testing systems
- Respect confidentiality and intellectual property
- Report vulnerabilities responsibly
- Consider harm to users and society

Case study: student portal

- Portal holds personal and academic data
- Identify privacy risks and access controls
- Discuss ethical use of logs and analytics

Practice questions

- Compare packet filtering, stateful inspection and proxies
- Explain firewall DMZ and limitations
- Differentiate copyright, patents and trademarks
- Discuss privacy versus security

Lecture Feedback



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